Notes on

Topology

Lecturers
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Jadavpur University

This text consists of notes on the course Topology, lectured by Professor Pratulananda Das (PD) and Professor Molla Basir Ahamed (MBA) at Jadavpur University in the 5th semester 2025.

Some changes and some additions have been made by me. To distinguish them from the lecture's actual contents, they are labelled with an asterisk. So any *Lemma** or *Remark** or *Proof** that the reader might come across are wholly my responsibility.

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List of Lectures

Course organisation. — The lectures will usually take place on Mondays from from 10:00 to 12:00 (MBA) and on Thursdays from 11:00 to 13:00, in the UG-I Exam Hall on the second floor of Ramanujan Bhavan.

For this course, the default recommendation is Munkres' *Topology* ([Mun17]).

The syllabus of this course is available here.

Chapter 1.

Cardinality

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1.1. Cardinality of sets

Lecture 1
7^{th} Aug, 2025

1.1.1. Preliminaries. — We want to compare the sizes of sets.

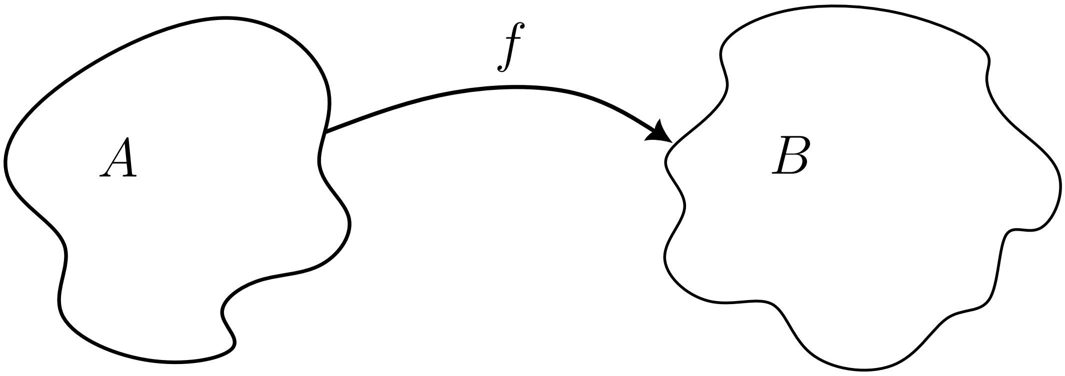

Figure 1.1.: Function from set A to B

To compare the sizes of A and B we consider mappings $f : A \to B$. We recall that

- f is **surjective** (or *onto*) if for any $b \in B$ there exists $a \in A$ such that $f(a) = b$.
- f is **injective** (or *one-one*) if the images of distinct elements from A are distinct.
- f is **bijective** if f is both surjective and injective.

1.1.2. Definition. — Two sets A and B are said to have the same cardinality, written as $|A| = |B|$, iff there exists a bijection from A onto B.

$|A| \leqslant |B|$ if there exists an injective map $f : A \to B$. To see this, simply consider the restricted map $f : A \to \operatorname{ran}(f) \subset B$ and denote it by $g : A \to \operatorname{ran}(f)$. Obviously g is bijective. Then $|A| = |\operatorname{ran}(f)| \leqslant |B|$ due to inclusion map.

1.1.3. Definition. — We say that $|A| \leqslant |B|$ iff there exists a bijection from A onto a subset B_1 of B.

1.1.4. Remark. — Clearly, cardinality with respect "$=$" is an equivalence relation and with respect to "$\leqslant$" is a partial order relation - although antisymmetry is not trivial.

1.1.5. Lemma. — *$|A| \leqslant |B|$ if there exists a surjection from B onto A.*

Proof. Suppose $f : B \to A$ is a surjection. Now for any $a \in A$ consider the set

$$f^{-1}(a) = \{b \in B : f(b) = a\}$$

called the **fibre** of a. Since f is onto, $f^{-1}(a) \neq \varnothing \quad \forall a \in A$. Also, if $a_1 \neq a_2$ then

$$f^{-1}(a_1) \cap f^{-1}(a_2) = \varnothing.$$

For, otherwise, if $f^{-1}(a_1) \cap f^{-1}(a_2) \neq \varnothing$ then we can choose some $b \in f^{-1}(a_1) \cap f^{-1}(a_2)$. But this implies that $f(b) = a_1$ and $f(b) = a_2$, which is absurd.

Hence, every fibre is nonempty and pairwise disjoint.

1.1.6. Axiom of Choice. — *Given a collection of pairwise disjoint nonempty sets, one can construct another set by taking one and only one element from each of these sets.* If the number of sets is finite, this is provable. If the number of sets is infinite, this can neither be proven nor disproven - hence it is an *axiom*.

Using the Axiom of Choice, now, we can construct a set $B_1 \sqsubset B$ taking one and only one element from each fibre $f^{-1}(a)$ for $a \in A$. Now observe that $f : B_1 \to A$ is a bijection. Then $f^{-1} : A \to B$ is also a bijection. $\square$

We previously noted that the antisymmetry of cardinality with respect to $\leqslant$ is not trivial. Indeed, it is the following theorem.

1.1.7. Theorem (Schröder-Bernsetein Theorem)**.** —

$$\textit{If } |A| \leqslant |B| \textit{ and } |B| \leqslant |A| \textit{ then } |A| = |B|.$$

1.1.8. Lemma. — *If $A_2 \subset A_1 \subset A$ and $|A| = |A_2|$ then $|A| = |A_1|$.*

Proof of lemma. Since $|A| = |A_2|$ there exists a bijection $f : A \to A_2$.

Then f induces a bijection from A_1 onto a set $A_3 \sqsubset A_2$ and then a bijection from A_2 onto a set $A_4 \sqsubset A_3$ and so forth. Consequently, we have a sequence of inclusions

$$A \supset A_1 \supset A_2 \supset A_3 \supset A_4 \supset A_5 \supset A_6 \supset A_7 \supset A_8 \supset \cdots$$

where $|A| = |A_2| = |A_4| = |A_6| = |A_8| = \cdots$ and $|A_1| = |A_3| = |A_5| = |A_7| = \cdots$.

Further, as f is a bijection, so we will also have (since bijection preserves set difference)

$$|A \setminus A_1| = |A_2 \setminus A_3| = |A_4 \setminus A_5| = \cdots.$$

Now let $D = \bigcap_{n=1}^{\infty} A_n$. Then

$$A = D \cup (A \setminus A_1) \cup (A_1 \setminus A_2) \cup (A_2 \setminus A_3) \cup (A_3 \setminus A_4) \cup (A_4 \setminus A_5) \cup \cdots$$
$$A_1 = D \cup (A_1 \setminus A_2) \cup (A_2 \setminus A_3) \cup (A_3 \setminus A_4) \cup (A_4 \setminus A_5) \cup (A_5 \setminus A_6) \cup \cdots$$

because every element a from A (or A_1) either belongs to A_n for all $n \in \mathbb{N}$ or, if not, then there exists an $n \in \mathbb{N}$ such that $a \in A_n$ but $a \notin A_{n+1}$. Now observe that

$$\Big[(A \setminus A_1) \cup (A_2 \setminus A_3) \cup (A_4 \setminus A_5) \cup \cdots\Big] \cong \Big[(A_2 \setminus A_3) \cup (A_4 \setminus A_5) \cup \cdots\Big]$$

which means

$$\Big|(A \setminus A_1) \cup (A_2 \setminus A_3) \cup (A_4 \setminus A_5) \cup \cdots\Big| = \Big|(A_2 \setminus A_3) \cup (A_4 \setminus A_5) \cup \cdots\Big|.$$

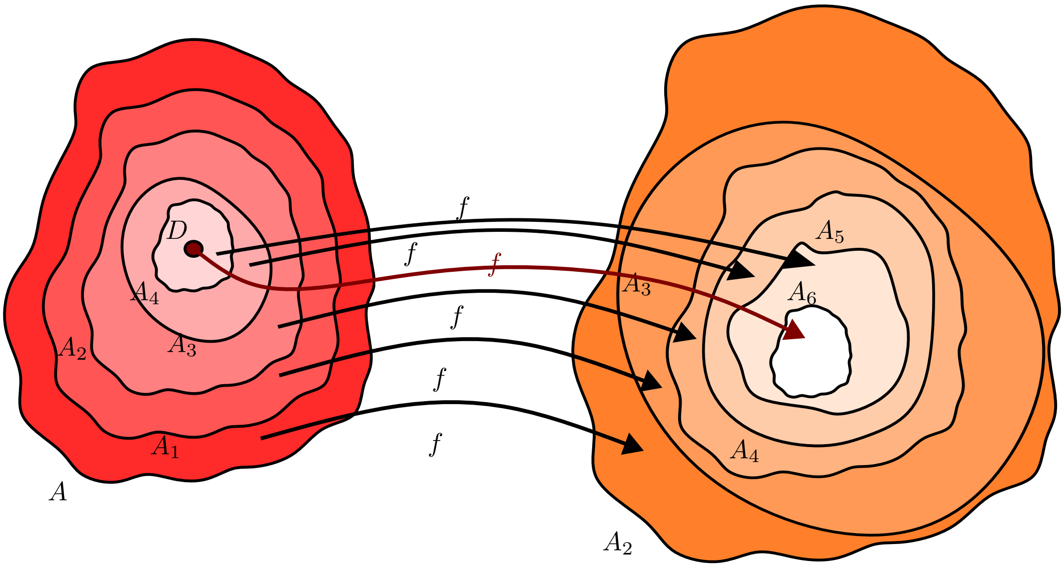

Figure 1.2.: $f(A) = A_2,\ A_1 \subset A,\ A_3 = f(A_1) \sqsubset f(A) = A_2$.

This means that there exists a bijection

$$g : \Big[(A \setminus A_1) \cup (A_2 \setminus A_3) \cup (A_4 \setminus A_5) \cup \cdots\Big] \to \Big[(A_2 \setminus A_3) \cup (A_4 \setminus A_5) \cup \cdots\Big].$$

We define $h : A \to A_1$ by

$$h(x) = \begin{cases} g(x), & \text{if } x \in (A \setminus A_1) \cup (A_2 \setminus A_3) \cup (A_4 \setminus A_5) \cup \cdots \\ x, & \text{if } x \in D \cup (A \setminus A_1) \cup (A_1 \setminus A_2) \cup (A_2 \setminus A_3) \cup \cdots . \end{cases}$$

Essentially, we pasted two bijections to get another bijection $h : A \to A_1$ where

$$A = (A \setminus A_1) \cup (A_1 \setminus A_2) \cup (A_2 \setminus A_3) \cup (A_3 \setminus A_4) \cup (A_4 \setminus A_5) \cup \cdots ,$$

$$A_1 = D \cup (A \setminus A_1) \cup (A_1 \setminus A_2) \cup (A_2 \setminus A_3) \cup (A_3 \setminus A_4) \cup (A_4 \setminus A_5) \cup \cdots .$$

Obviously, h is a bijection from A onto A_1 and hence we conclude $|A| = |A_1|$. □

Proof of Theorem 1.1.7 (Schröder-Bernsetein). Let $|A| \leqslant |B|$ and $|B| \leqslant |A|$.

Then $|A| = |B_1|$ where $B_1 \sqsubset B$ and $|B| = |A_1|$ where $A_1 \sqsubset A$. But then $A_1 \subset A$ and there exists a bijection $f : A \to B_1$ implying that there exists another bijection from A_1 onto $B_2 \sqsubset B_1$. Then $|B_2| = |A_1| = |B|$. Now $B_2 \sqsubset B_1 \sqsubset B$ and $|B_2| = |B| \implies |B_1| = |B|$. Hence, $|A| = |B|$. □

1.1.9. Countable sets. — As an application of 1.1.7 we'll show that $|\mathbb{Z}| = |\mathbb{N}|$. First notice that $\mathbb{N} \subset \mathbb{Z}$ so of course $|\mathbb{N}| \leqslant |\mathbb{Z}|$. Now, for any $n \in \mathbb{Z}$ define

$$f(n) = \begin{cases} 2n+1, & \text{if } n \geqslant 0 \\ 2|n|, & \text{if } n < 0. \end{cases}$$

where the choice of 2 makes f a bijection; but we could also choose 4 instead of 2

$$f(n) = \begin{cases} 4n+1, & \text{if } n \geqslant 0 \\ 4|n|, & \text{if } n < 0. \end{cases}$$

so that now f is only injective, but this is sufficient for us to conclude that $|\mathbb{Z}| \leqslant |\mathbb{N}|$.

1.1.10. Definition. — A set A is **countable** iff $|A| = |\mathbb{N}|$.

1.1.11. Lemma. — *If A and B are countable, then $A \cup B$ and $A \times B$ are also countable.*

Proof. There exist bijections $f : A \to \mathbb{N}$ and $g : B \to \mathbb{N}$. Define $h : A \cup B \to \mathbb{N}$ by

$$h(x) = \begin{cases} 2f(a), & \text{if } x = a \in A \\ 2g(b)+1, & \text{if } x = b \in B. \end{cases}$$

then h is injective so $A \cup B$ is countable.

Also $(a, b) \mapsto 2^{f(a)}3^{g(b)}$ defines an injection from $A \times B$ to $\mathbb{N}$. □

As a result, $\mathbb{N} \times \mathbb{N}$ and $\mathbb{Q}$ are also countable. For $\mathbb{Q}$, we can map any $\frac{m}{n} \in \mathbb{Q}$ to $(2m+1, n)$ for $m \geqslant 0$ and $(2|m|, n)$ for $m < 0$ to get an injection from $\mathbb{Q}$ into $\mathbb{N} \times \mathbb{N}$.

Lecture 2
11th Aug, 2025

Even though today is Monday, PD is taking the class instead of MBA because PD won't be coming next Thursday (or even next Monday for that matter). Returning to the lecture, by default we will assume all pairs or collections of sets to be pairwise disjoint - even if we don't write it down explicitly all the time, we will still assume this subconsciously.

Now consider a sequence of countable sets $\{A_n\}_n$, then we want to show $\bigcup_n A_n$ is countable.

First, observe that if we want to make two (disjoint) copies copies of a set A, we can do the following

$$\begin{tikzcd} & A \arrow[dl] \arrow[dr] & \\ A \times \{0\} & & A \times \{1\} \end{tikzcd}$$

where $A \times \{0\} = \{(a, 0) : a \in A\}$ and $A \times \{1\} = \{(a, 1) : a \in A\}$. The sets are disjoint as they differ in the 2nd coordinate for each element and we also have the equivalence: $A \cong A \times \{0\} \cong A \times \{1\}$.

For a countable number of pairwise disjoint sets, each equivalent to A, we can construct

$$A \times \{n\} : n \in \mathbb{N}.$$

For the uncountable case, we can take $A \times \{\alpha\} : \alpha \in (0, 1)$ where we use the fact that $(0, 1)$ is an uncountable set. Now, as there are infinitely many primes, we can always construct a sequence of primes.

Let $\{p_n\}_n$ be a sequence of primes and let A_n be countable. Then $A_n \cong \mathbb{N}$, which means that there exists a bijection $f_n : A_n \to \mathbb{N}$. Now define $H : \bigcup_n A_n \to \mathbb{N}$ by

$$h(a) = p_n^{f_n(a)} \quad \text{if } a \in A_n.$$ [1]

Obviously h is injective so

$$\left| \bigcup_n A_n \right| \leqslant |\mathbb{N}|$$

and so $\bigcup_n A_n$ is countable.

1.1.12. Finite and infinite sets. — We need one more set-theoretic result:

1.1.13. Definition. — A set A is **finite** if there exists an injective map from A to an initial segment of $\mathbb{N}$ i.e. $\{1, 2, \ldots, k\}$ for some $k \in \mathbb{N}$. If the set A is not finite then it is said to be **infinite**.

This means given any length-k initial segment of $\mathbb{N}$, if A is infinite then for any f mapping A to the segment there will always be an image of f outside of the segment.

1.1.14. Proposition. — *If A is an infinite set, then $|A| \leqslant |\mathbb{N}|$.*

Proof. Take $a_1 \in A$. Since A is infinite we can choose $a_2 \in A \setminus \{a_1\}$.

$$\vdots$$

For any n we can choose $a_n \in A \setminus \{a_1, a_2, \ldots, a_{n-1}\}$.

This implies that we can choose a sequence $C = \{a_n\}_n \sqsubset A$. Obviously, $C \cong \mathbb{N}$ by the mapping $h(n) = a_n$, so $h : \mathbb{N} \to A$ is injective and so $|\mathbb{N}| \leqslant |A|$. $\square$

1.1.15. Theorem. — *For any two sets A and B, either $|A| \leqslant |B|$ or $|B| \leqslant |A|$.*

Proof. If A and B are finite or if at least one of them is finite then the proof is trivial.[2] So without any loss of generality (WLOG), we assume that both A and B are infinite. We want to show an injection from either A to B or B to A. As we know nothing about A and B, we cannot proceed in our usual way by constructing explicit maps; we look at their subsets instead.

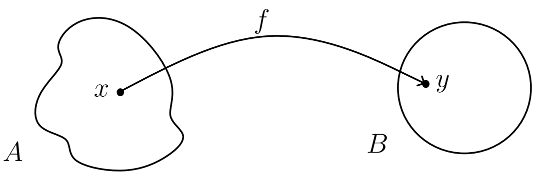

Figure 1.3.: Such $x \in A$ and $y \in B$ always exist as both are infinite sets. We map singletons $\{x\}$ to $\{y\}$ and get trivial bijections.

Let $\mathbb{F}$ be the collection of all bijections f with $\mathrm{dom}(f) \subset A$ and $\mathrm{ran}(f) \subset B$. Then $\mathbb{F} \neq \varnothing$. Now for any two mappings $f_1, f_2 \in \mathbb{F}$ let us define a relation "$\sqsubset$" by $f_1 \subset f_2$ if f_2 is an extension of f_1, i.e. $\mathrm{dom}(f_1) \sqsubset \mathrm{dom}(f_2)$, $\mathrm{ran}(f_1) \sqsubset \mathrm{ran}(f_2)$.

[1] Again, we don't need bijection; an injection suffices.

[2] In the first case since $A \cong \{1, \ldots, k\}$ and $B \cong \{1, \ldots, m\}$ and either $k \leqslant m$ or $m \leqslant k$. In the second case, if (say) B is infinite then $|A| \leqslant |B|$.

Now $\mathbb{F}$ is a poset with respect to the relation "$\sqsubset$" (reflexive, transitive, antisymmetric). If $C \subset \mathbb{F}$ is a chain then observe that $h = \bigcup_{f \in C} f$ is an upper bound of C. Here we define

$$\operatorname{dom}(h) = \bigcup_{f \in C} \operatorname{dom}(f), \ \operatorname{ran}(h) = \bigcup_{f \in C} \operatorname{ran}(f).$$

$\operatorname{dom}(h) \subset A, \operatorname{ran}(h) \subset B$ and C chain $\implies$ preimages of h are unique. So h is well-defined.

Then $\operatorname{dom}(h) \sqsubset A$, $\operatorname{ran}(h) \sqsubset B$ and $h : \operatorname{dom}(h) \to \operatorname{ran}(h)$ is clearly a bijection.

As h is well-defined, $x \in \operatorname{dom}(h) \implies x \in \operatorname{dom}(f)$ for some $f \in C$. So $h(x) = f(x)$. Also, if $f \leqslant g$ then $g(x) = f(x)$. If we take $x \neq y$ in $\operatorname{dom}(h)$ then $x \in \operatorname{dom}(f_1)$ and $y \in \operatorname{dom}(f_2)$, $f_1, f_2 \in C$ (a chain). WLOG let $f_1 \leqslant f_2$, then $x, y \in \operatorname{dom}(f_2)$, but f_2 is a bijection

$$h(x) = f_2(x) \neq f_2(y) = h(y).$$

So $(\mathbb{F}, \sqsubset)$ is a poset where every chain has an upper bound. We now invoke Zorn's Lemma.

1.1.16. Zorn's Lemma. — If in a poset $(P, \leqslant)$ every chain has an upper bound then P has a **maximal**[3] element. [i.e. $\exists p_0 \in P : p_0 \not\leqslant p$ for any $p \in P : p \neq p_0$]

By Zorn's Lemma, there exists a maximal element g of $\mathbb{F}$. Clearly, g is a bijective map with $\operatorname{dom}(g) \sqsubset A, \operatorname{ran}(g) \sqsubset B$.

If $\operatorname{dom}(g) = A$, then clearly $|A| \leqslant |B|$. If $\operatorname{ran}(g) = B$, then $g^{-1} : B \to \operatorname{dom}(g) \sqsubset A$ is an injective map so $|B| \leqslant |A|$.

If not any of the above two cases, $A \setminus \operatorname{dom}(g) \neq \varnothing \neq B \setminus \operatorname{ran}(g)$. Take $x \in A \setminus \operatorname{dom}(g)$

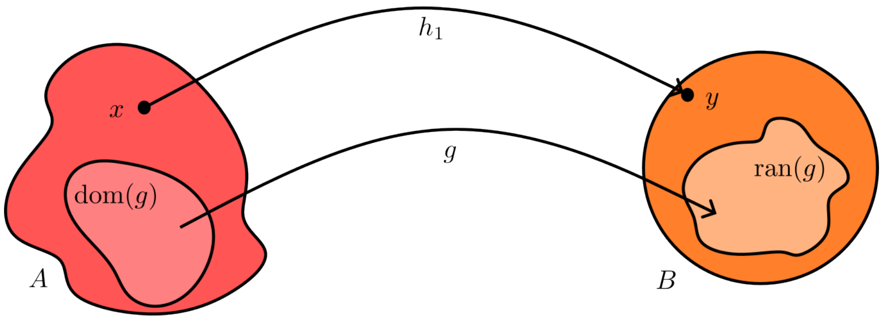

Figure 1.4.: $g : \operatorname{dom}(g) \to \operatorname{ran}(g)$.

and $y \in B \setminus \operatorname{ran}(g)$ and let $h_1(x) = y$. Then

$$h = h_1 \cup g : \operatorname{dom}(g) \cup \{x\} \to \operatorname{ran}(g) \cup \{y\}$$

given by $h(x) = g(x)$ if $a \in \operatorname{dom}(g)$; $h(x) = h_1(x) = y$ is a bijection and so $h \in \mathbb{F}$ and $g \sqsubset h$ where $h \neq g$ contradicting the maximality of g. $\square$

1.1.17. Remark. — We already know cardinality with respect to $\leqslant$ is a partial order relation. This theorem shows that it is in fact a total order. A totally ordered set is called a **chain**.

[3] no element greater than or equal to it; different from *maximum* which every element is less than or equal to

1.1.18. Definition. — We write $|A| < |B|$ if $|A| \leqslant |B|$ and $|A| \neq |B|$.

1.1.19. Theorem. — *For any set A, $|A| < |\mathcal{P}(A)|$.*

1.1.20. Remark. — This means there is no maximal set with respect to the cardinality relation. Popularly this result is known as "infinity of infinities".

Cantor's proof. $|A| \leqslant |\mathcal{P}(A)|$ is clear from the mapping $f : A \to \mathcal{P}(A)$ defined by $f(x) = \{x\}$ for each $x \in A$ and in view of the fact that f is injective. In order to show that $|A| < |\mathcal{P}(A)|$ we start by assuming towards a contradction (this seems to be the only viable approach) that $|A| = |\mathcal{P}(A)|$ instead.

Then $\exists$ a bijection $h : A \to \mathcal{P}(A)$. Consider the set (possibly empty but well-defined)

$$C = \{x \in A : x \notin h(x)\}$$

then clearly $C \sqsubset A$ and so $C \in \mathcal{P}(A)$. Since h is surjective, $\exists p \in A : h(p) = C$. Now, either $p \in C$ or $p \notin C$.

If $p \in C$ then $p \notin h(p) = C$, which is an absurdity. On the contrary, if $p \notin C$, then $p \notin h(p)$ implies that $p \in C = h(p)$ by the very construction of C, which is again an absurdity.[4]

Hence, $|A| \neq |\mathcal{P}(A)|$ and so $|A| < \mathcal{P}(A)$. □

So given any set we can always find a bigger set, even if the set is infinite. Here, the very natural numbers provide us with intuition. $\mathbb{N}$ itself is a chain with no maximal element. Similarly, the collection of sets under the cardinality relation is a chain that has no maximal element. This why later on we will introduce the notion of 'cardinal *numbers*' as well as 'ordinal *numbers*'.

1.1.21. Theorem. — *For any two sets A, B*

$$|A| \leqslant |A \cup B|.$$

Proof. Obvious. (So homework !) □

1.1.22. Theorem. —

$$|A \cup \{y_1, y_2, \ldots, y_n\}| = A$$

for any infinite set A where $A \cap \{y_1, y_2, \ldots, y_n\} = \varnothing$.

Proof. As A is infinite, we can choose a sequence of distinct elements in A i.e. $\exists$ a countable set $\{x_1, x_2, \ldots, x_n, \ldots\} \sqsubset A$. Let us write $C = \{x_n\}_n$. First, observe that we can partition A into C and $A \setminus C$ i.e. $A = C \cup (A \setminus C)$ where the union is of two disjoint sets.

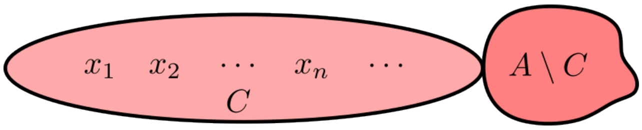

Figure 1.5.: We partition A into C and $A \setminus C$.

Therefore to show a bijection we only need to show one between C and $C \cup \{y_1, y_2, \ldots, y_n\}$.

The diagram above shows the bijection. We always put the finite part first. Homework: write down the map explicitly and complete the proof. □

[4] this style of argument reminds me of Russell's paradox

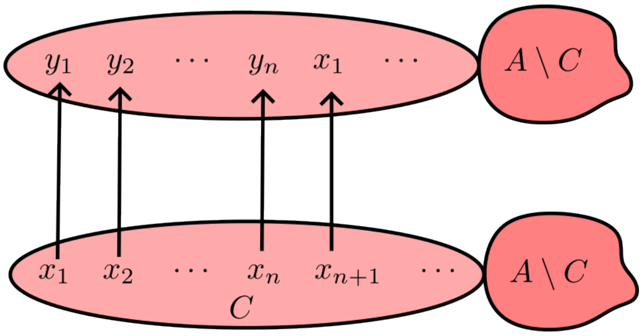

Figure 1.6.: Bijection between C and $C \cup \{y_1, y_2, \dots, y_n\}$.

1.1.23. Theorem. —

$$|A \cup \mathbb{N}| = |A|$$

where A is an infinite set.

Sketch of proof. Again let $C = \{x_1, x_2, \dots, x_n, \dots\} \sqsubset A$. We then show a bijection from $\mathbb{N} \cup A$ to A in 3 parts: show a bijection from $\mathbb{N}$ to $\{x_{2n}\}_n$, a bijection from C to $\{x_{2n-1}\}_n$ and then only $A \setminus C$ remains which maps to itself.

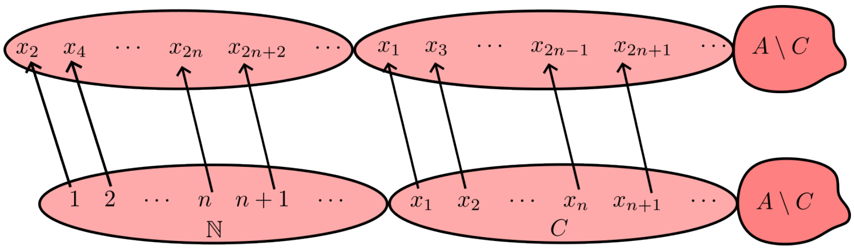

Figure 1.7.: Bijection between $\mathbb{N}$ and $\mathbb{N} \cup A$.

So we have a bijection from A to $\mathbb{N} \cup A$. □

In fact, taking even countable union of $\mathbb{N}$'s yields the same result. This is where our intuition for cardinality diverges from that of the natural numbers: $m + n > m$ is always true for any positive integer m, n. But for "cardinal numbers", this won't hold as we've seen in Theorems 1.1.22 and 1.1.23.

Bibliography

[Mun17] James R. Munkres. *Topology*. Pearson, 2017.